
Assignment 1: Optimization in Two Neurons Visualization

Deep Learning 2026

Due: ~2026.09.30 (Wed)

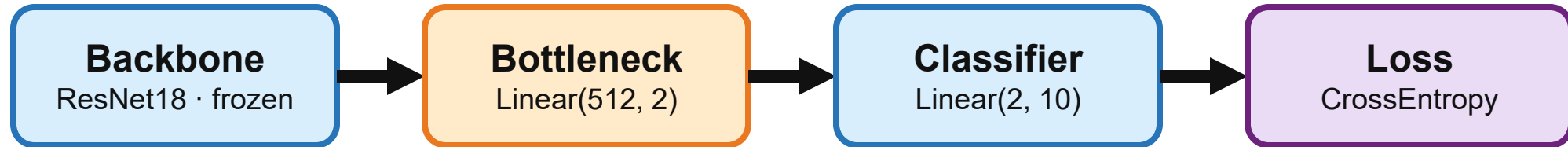
Joo Yong Sim



The Assignment 1: What to Build, What to Report



Compress a trained CIFAR-10 classifier to **two** neurons, then watch optimization work



1. Build the model : ResNet18 backbone (frozen) $\rightarrow$ Linear(512, 2) $\rightarrow$ Linear(2, 10), cross entropy

2. Measure what it costs : compare test accuracy, $512 \rightarrow 10$ against $512 \rightarrow 2 \rightarrow 10$

3. Visualize the 2-D representation : plot z over the test set — what do the two neurons encode?

4. Visualize the optimization : plot $L(w_1, w_2)$ and the descent path for several η

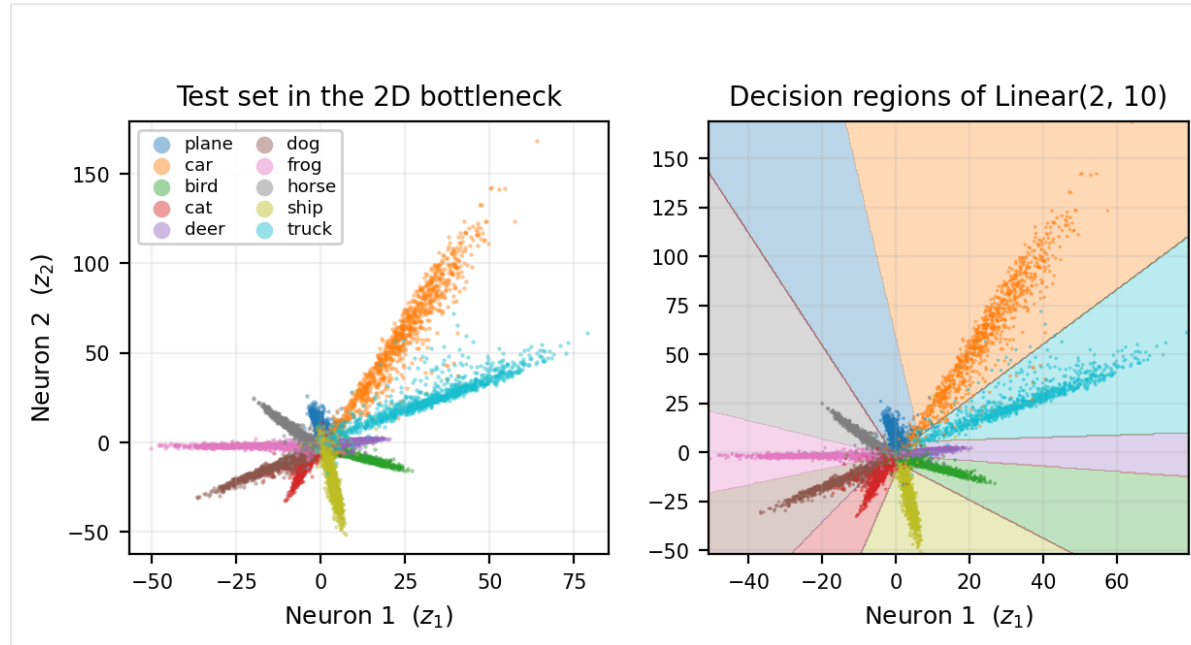
Deliverable : one figure per task, and one sentence saying what each figure shows

Tasks 2 – 3: The Cost of Two Neurons



At how much cost Ten classes survive a two-dimensional bottleneck with 2 Neurons?

Example:



10,000 test images · decision regions of Linear(2, 10) · per-class angle

Why two suffice : class identity rides on the **angle** — the radius only carries confidence

Setup : backbone fine-tuned 3 epochs on CIFAR-10 (64×64 input); head trained 40 epochs, Adam 1e-3

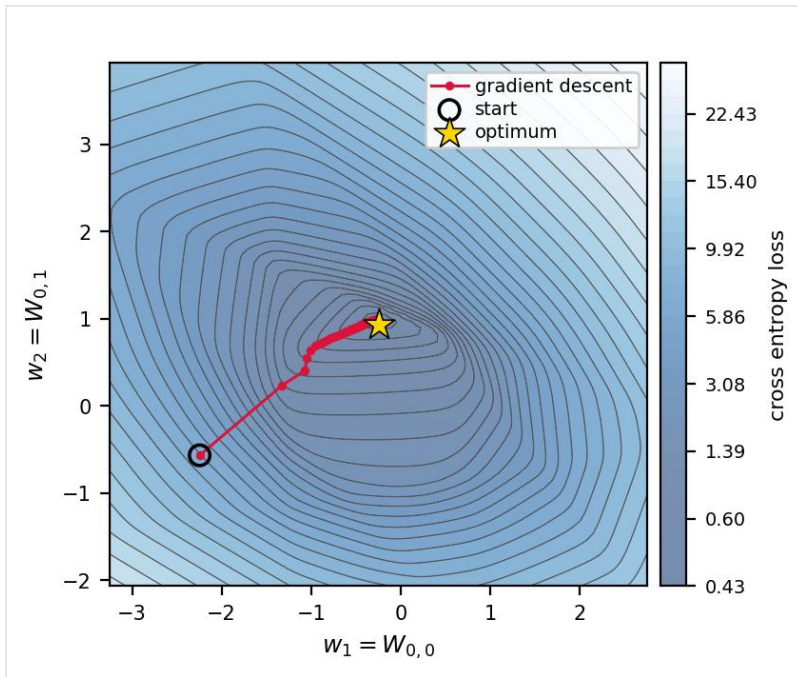
Where it overlaps : cat–dog and deer–horse — the pairs the full classifier already confuses

Task 4: Loss Landscape and Learning Rate



Two weights, not two neurons — $L(w_1, w_2)$ is a flat slice through a 1,056-parameter head

Example:



80 × 80 grid over one fixed batch of 128

